Graphic Era (Deemed to be University)

GEHU/02E/9.1.3

End Term (Even) Semester Regular/Back Examination May 2026

Roll no..2418159.............

Name of the Program and semester: B. Tech IVth Semester.
Name of the Course: Design and Analysis of Algorithms
Course Code: TCS 409

Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) – CO-1

a. Solve the following recurrence relation
   (i) $T(n) = 4T(n/2) + n^2 \log n$. - using master theorem
   (ii) $T(n) = T(n-1) + \log n$ -using substitution method

b. In algorithm design, the Tower of Hanoi problem is a classic example used to illustrate recursion, where a set of disks must be moved from a source peg to a destination peg specific rules.
   (i) Derive a recursive algorithm to solve the Tower of Hanoi problem for n disks.
   (ii) Determine the total number of function calls (or moves) required to solve the problem using the recursive approach. Also, analyze the time and space complexity of the recursive solution.

c. Write an algorithm for Exponential Search. Analyze their time complexities (best, average, and worst cases), and analyze under what conditions Linear, Binary and Exponential search techniques are most efficient.

Q2. (2X10=20 Marks)– CO-2

a. Write the pseudocode for Insertion Sort and explain its working for the given array: A = {6, 4, 2, 2, 8, 3, 5, 1} also justify insertion sort holds the online, stable and in-place properties.

b. Design a Backtracking algorithm for the Subset Sum problem. Given a set {4, 7, 9, 11, 13, 17} and a target sum = 26, trace the execution of the algorithm and list all possible subsets that satisfy the condition.

c. Write the pseudocode for the Quick Sort algorithm and explain its working for the array A = {8, 2, 4, 3, 9, 6, 10. Also, explain how Randomized Quick Sort modifies the pivot selection process and discusses how it helps in improving performance compared to standard Quick Sort.

Q3. (2X10=20 Marks)– CO-3

a. Write the pseudocode for Breadth-First Search (BFS) and analyze its time and space complexity.

b. Write a pseudocode for topological sort and analyze its time and space complexity.

c. Write the pseudocode for the Floyd–Warshall algorithm to compute the shortest paths between all pairs of vertices in a given below weighted graph and Compute the shortest distance and path matrix step-by-step.

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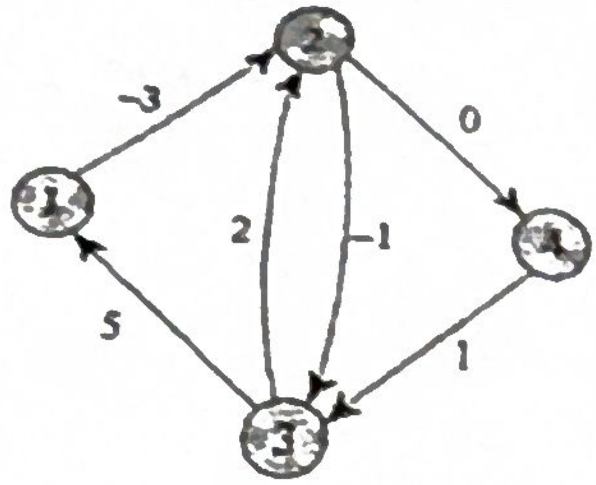

Q4.

(2X10=20 Marks) – CO-4

a. Write an algorithm (pseudocode) to compute the Longest Common Subsequence (LCS) of two given strings and analyze its time complexity. Given the strings: X = "AGGTAB" and Y = "GXTXAYB".

b. A communication system currently uses 7-bit fixed-length encoding and is being upgraded to Huffman Coding based on symbol probabilities:

M1: 0.45 M2: 0.02 M3: 0.24 M4: 0.18 M5: 0.11

(i) Construct the Huffman Tree and determine the codes.

(ii) Decode the message using the generated codes.

(iii) Calculate the average number of bits per symbol using Huffman coding.

c. Given a set of jobs where each job has a deadline and a profit as shown below:

Job	J1	J2	J3	J4	J5	J6
Profit	24	18	22	10	12	30
Deadline	5	3	3	2	4	2

(i). Determine the optimal sequence of jobs to maximize total profit.

(ii). Calculate the maximum profit obtained.

(iii). Analyze time complexity of the algorithm.

Q5.

(2X10=20 Marks) – CO-5

a. Explain any three commonly used hash functions with suitable examples for handling string keys. A hash function is defined as: H(key)=(sum of ASCII values of characters in the key) mod9 and linear probing is used for collision resolution. Insert the following string keys into a hash table indexed from 0 to 6:

"SUN", "MOON", "STAR", "SKY", "CLOUD", "RAIN", "WIND"

(i) Determine the final position of the key "CLOUD" in the hash table.

(ii) Count the total number of collisions that occur during the insertion process and justify your answer.

b. Describe the complexity classes P, NP, and NP-Complete, and explain the relationships among these classes, including how they are interconnected.

c. Compute the prefix function π for the pattern P = abacab using KNUTH-MORRIS-PRATT algorithm